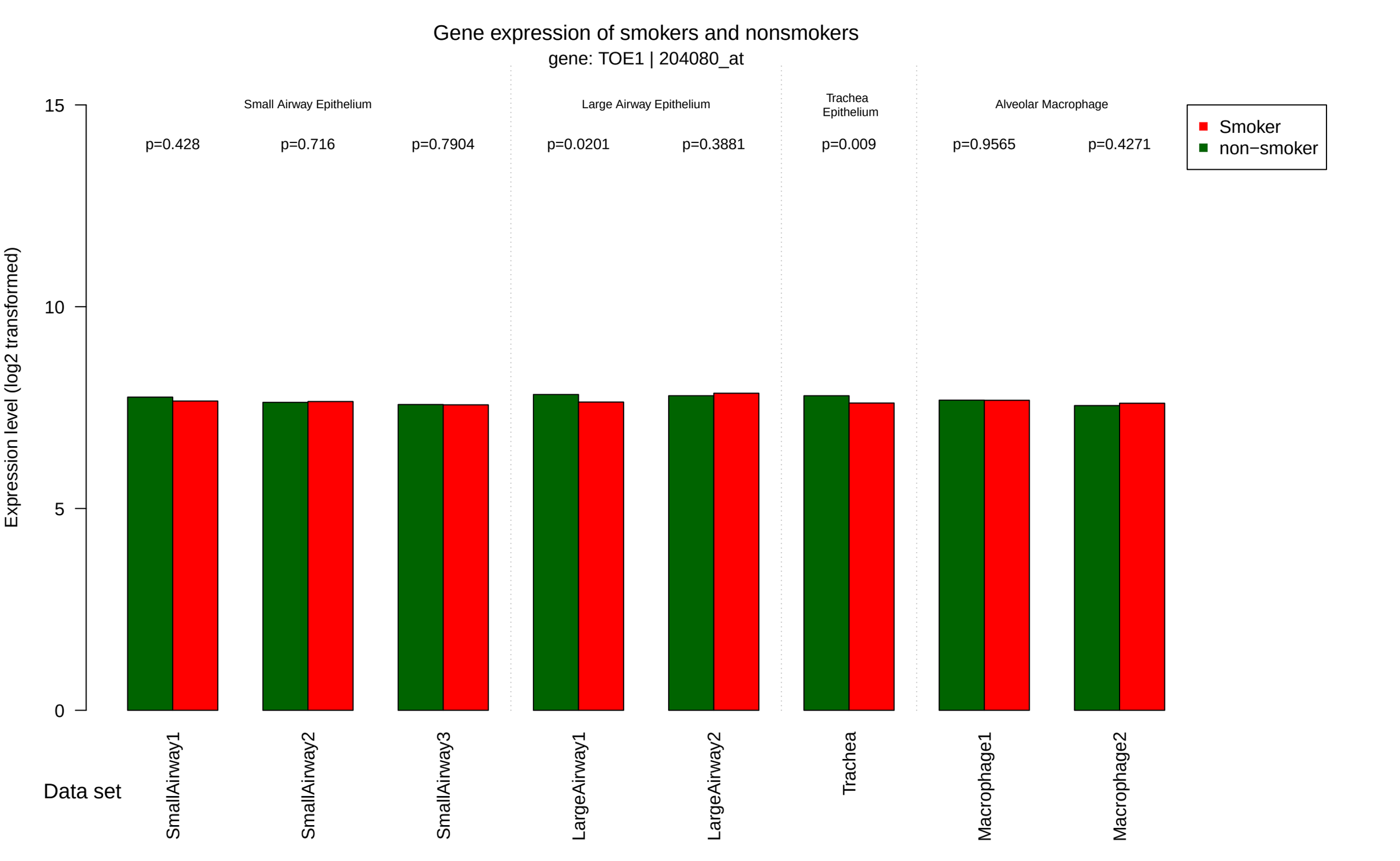
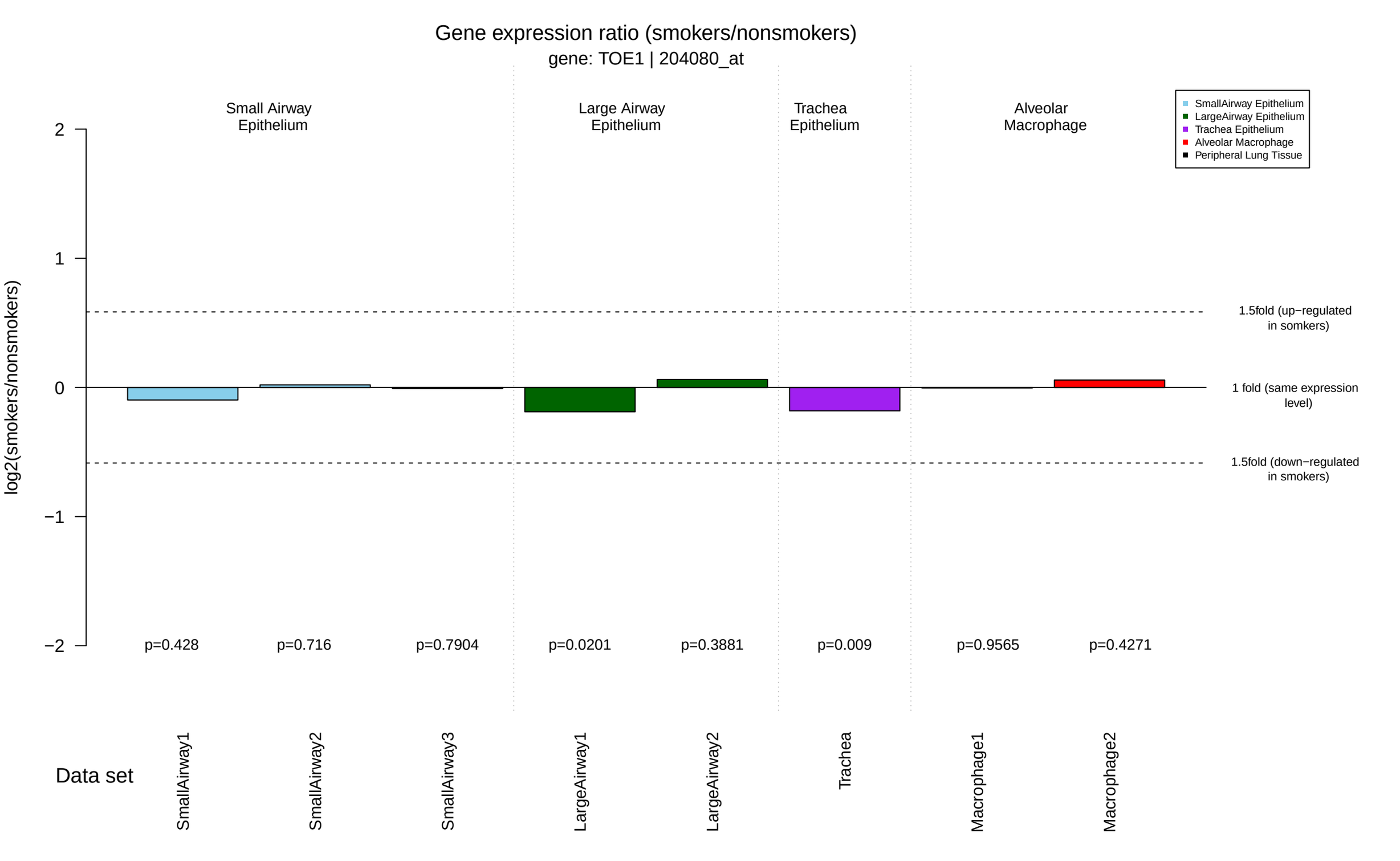
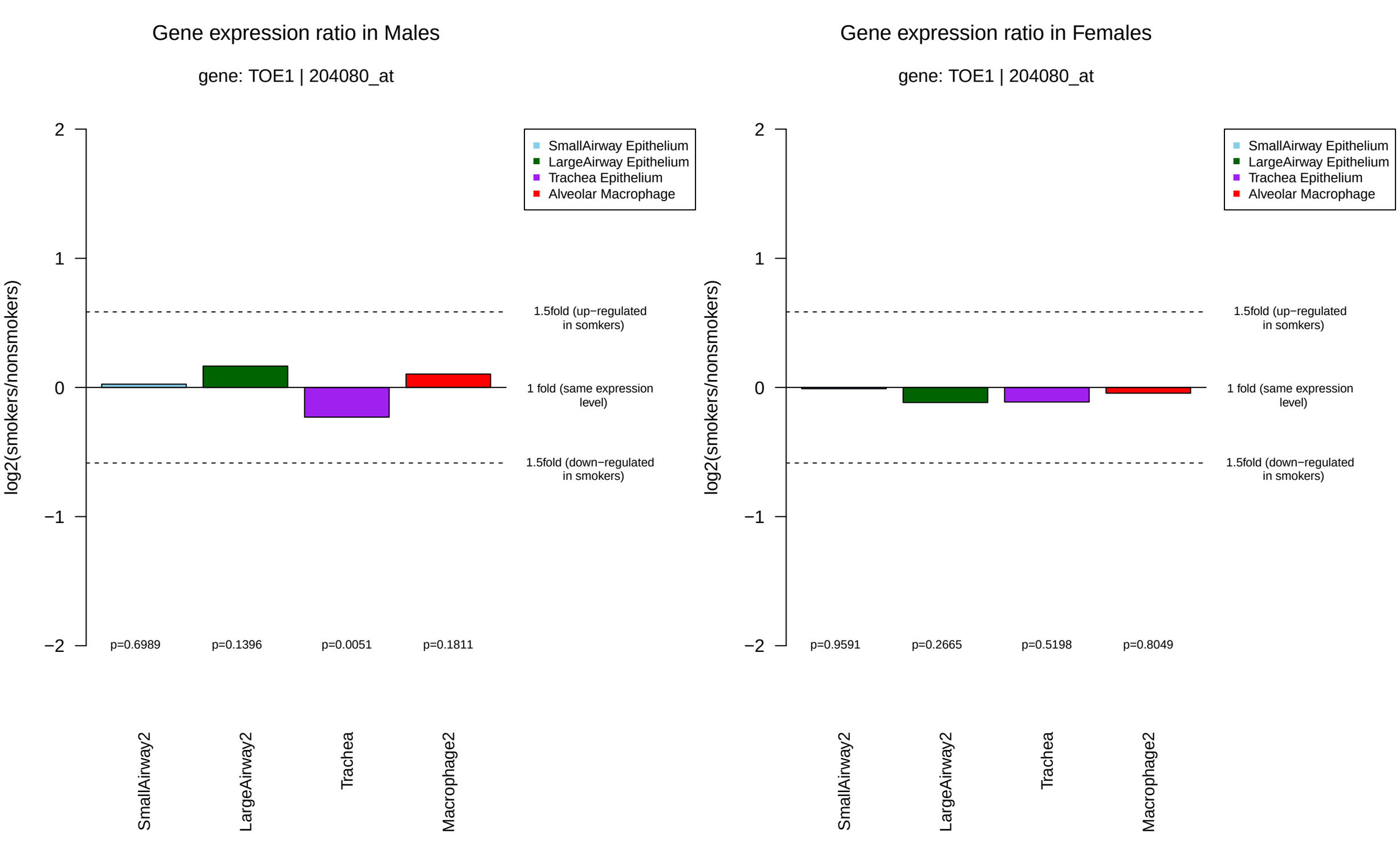
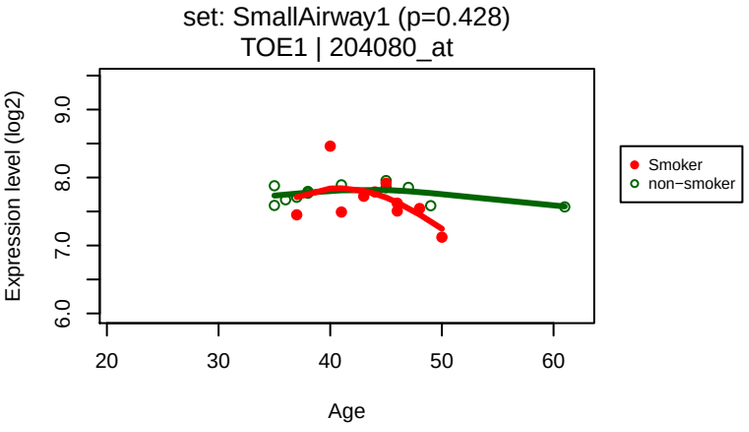
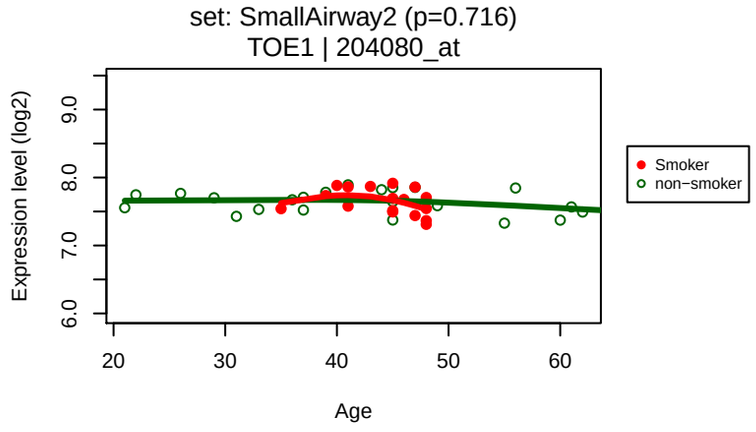
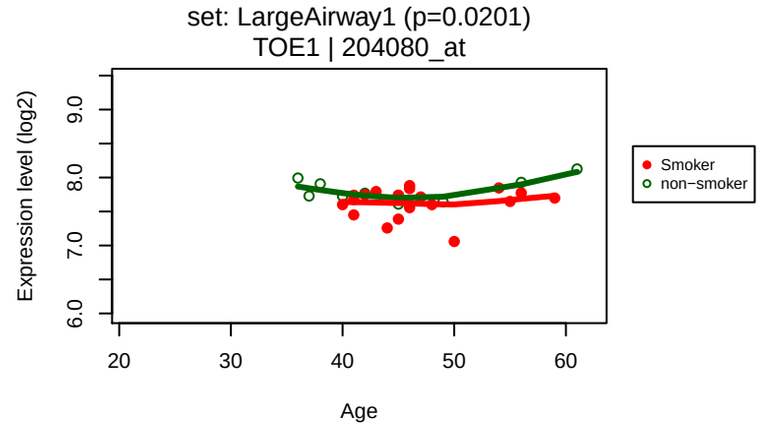

Gene expression of smokers and nonsmokers
gene: TOE1 | 204080_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.428
p=0.716
p=0.7904
p=0.0201
p=0.3881
p=0.009
p=0.9565
p=0.4271
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

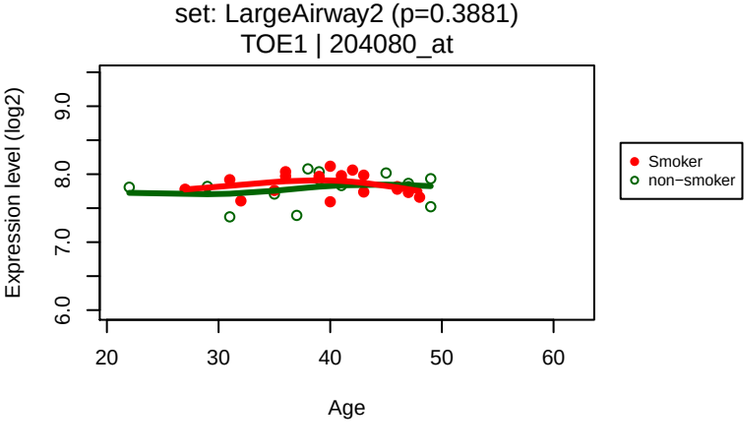

Gene expression ratio (smokers/nonsmokers)
gene: TOE1 | 204080_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.428
p=0.716
p=0.7904
p=0.0201
p=0.3881
p=0.009
p=0.9565
p=0.4271
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

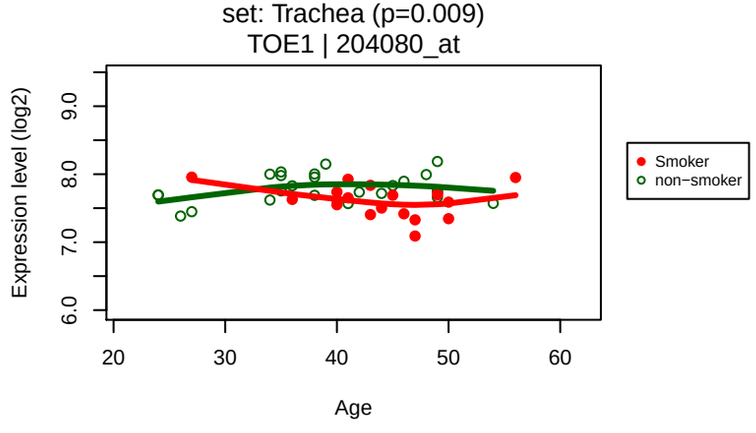

Gene expression ratio in Males
gene: TOE1 | 204080_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6989
p=0.1396
p=0.0051
p=0.1811
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TOE1 | 204080_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9591
p=0.2665
p=0.5198
p=0.8049
SmallAirway2
LargeAirway2
Trachea
Macrophage2

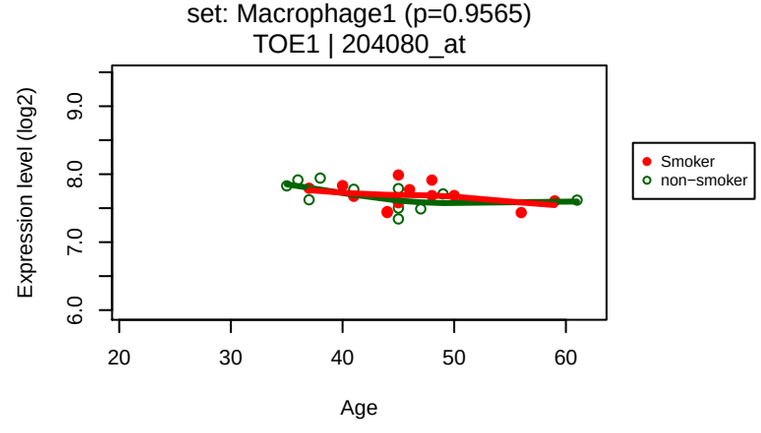

set: SmallAirway1 (p=0.428)
TOE1 | 204080_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker

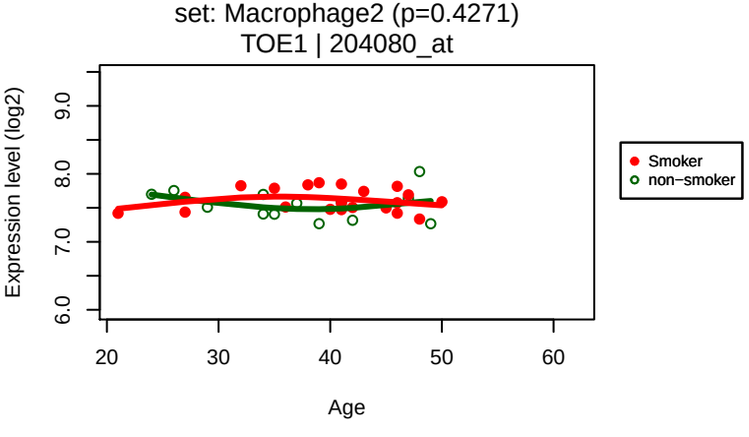

set: SmallAirway2 (p=0.716)
TOE1 | 204080_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker

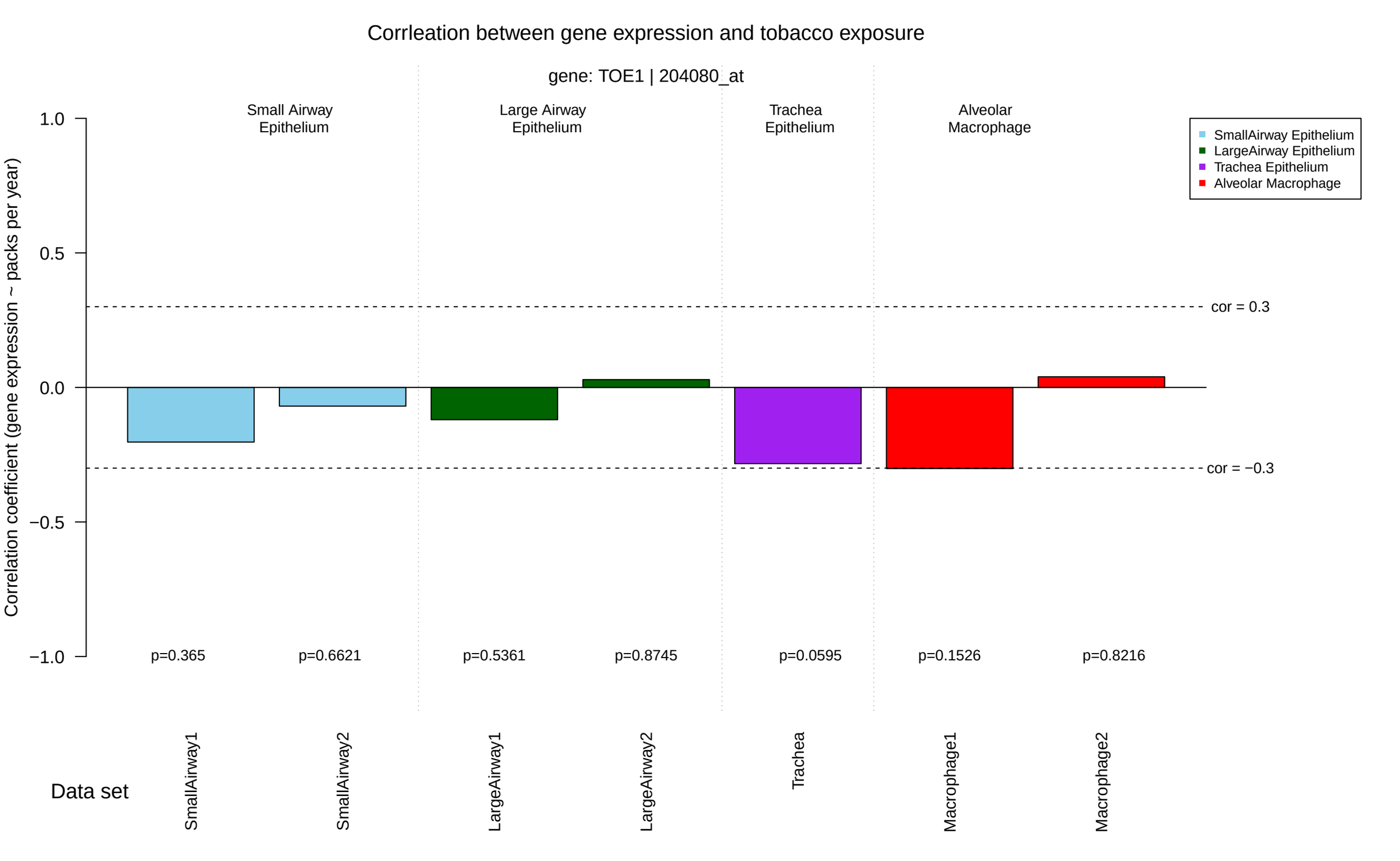

set: LargeAirway1 (p=0.0201)
TOE1 | 204080_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3881)
TOE1 | 204080_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.009)
TOE1 | 204080_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9565)
TOE1 | 204080_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4271)
TOE1 | 204080_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TOE1 | 204080_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.365
p=0.6621
p=0.5361
p=0.8745
p=0.0595
p=0.1526
p=0.8216
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2